

Sruthi Satyavarapu

AI/ML Engineer | Generative AI | RAG & Agentic AI | MLOps

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PROFESSIONAL SUMMARY

AI/ML Engineer with 7+ years of experience spanning machine learning, backend engineering, and cloud-native systems, with specialization in Generative AI and LLM applications. Experienced in designing and deploying production-grade RAG and Agentic AI systems, ML pipelines, and model-serving infrastructure using Python, PyTorch, AWS, Kubernetes, Docker, and SageMaker. Strong expertise in retrieval optimization, LLM evaluation, fine-tuning, vector search, and MLOps, with research experience in modular RAG architectures and domain adaptation. Published IEEE research on domain-adaptive RAG systems and demonstrated impact through measurable improvements in ML accuracy, processing efficiency, and system latency.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C++, Java, JavaScript, TypeScript
- **Generative AI & LLMs:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, AI Agents, Multi-Agent Systems, Prompt Engineering, LangChain, LlamaIndex, LangGraph, Hugging Face, Anthropic Claude API, OpenAI API, AWS Bedrock, LoRA, Fine-Tuning
- **RAG Evaluation & Retrieval:** RAGAS, Recall@K, MRR, Cross-Encoder Re-Ranking, Query Rewriting, Hybrid Retrieval, Chunking Strategies, LangSmith, Component-Level Latency Instrumentation
- **Machine Learning & AI:** PyTorch, Scikit-learn, XGBoost, NLP, Deep Learning, Time-Series Forecasting, Anomaly Detection
- **MLOps & Model Deployment:** AWS SageMaker, MLflow, Docker, Kubernetes, Apache Airflow, CI/CD, Model Monitoring, Model Drift Detection, Experiment Tracking, Model Serving
- **Cloud Platforms:** AWS (EC2, S3, Lambda, SageMaker, Bedrock, IAM), Cloud Deployment, Distributed Systems
- **Databases & Vector Search:** PostgreSQL, MongoDB, MySQL, ChromaDB, Pinecone, FAISS, Weaviate, Embedding Pipelines, Semantic Search, Hybrid Search
- **Backend Engineering:** FastAPI, Flask, Spring Boot, REST APIs, Microservices, ETL Pipelines, Pandas, NumPy, Apache Spark
- **AI System Security & Access:** RBAC, OAuth2, JWT, OWASP, Multi-Tenant Access Control, Session Management

PUBLICATIONS

S. Satyavarapu, P. Gupta, M. Fahad. "A Service-Oriented Modular Architecture for Domain-Adaptive Retrieval-Augmented Generation." *IEEE International Conference on Big Data Service and Applications (BigDataService)*, Fukuoka, Japan, 2026.

DOI: [10.1109/BigDataService70481.2026.00047](https://doi.org/10.1109/BigDataService70481.2026.00047)

EDUCATION

M.S. Computer Science | University of the Pacific, Stockton, CA

Aug 2024 – June 2026 | GPA: 4.0

M.S. Technology (Medical Software) | Manipal Institute of Technology, India

Aug 2014 – May 2016 | GPA: 3.7

B.Tech (BioMedical Engineering) | Jawaharlal Nehru Technological University, India

Aug 2010 – May 2014 | GPA: 3.6

RESEARCH EXPERIENCE

Research Assistant | RAG & LLM Systems | University of the Pacific

Mar 2025 – Present • Stockton, CA

- Architected a service-oriented, modular RAG system enabling component swapping (embedding models, retrievers, vector stores, LLM backends) through configuration alone, bridging benchmarking frameworks (FlashRAG) and production frameworks (LangChain, LlamaIndex).
- Built the end-to-end system in ReactJS, FastAPI, and ChromaDB with query rewriting, cross-encoder re-ranking, session-aware multi-turn interaction, and component-level latency instrumentation, with PostgreSQL backing session history, RBAC policies, and per-user pipeline configuration for multi-user deployment.
- Evaluated across biomedical and Sanskrit philosophical corpora using Recall@K, MRR, and RAGAS, showing identical configurations behave very differently by domain.
- Currently extending the framework with LoRA-based parameter-efficient fine-tuning to compare configuration-driven domain adaptation against fine-tuned retrieval within the same modular pipeline.

Research Assistant | ML for IoT & Health Systems | University of the Pacific

Jun 2025 – Dec 2025 • Stockton, CA

- Developed anomaly detection models using PyTorch and Scikit-learn on wearable sensor data streams for IoT-based health monitoring, improving detection accuracy by 28% over baseline.
- Engineered time-series feature extraction pipelines for environmental and motion-based sensor datasets, enhancing predictive model performance across multiple sensor modalities.
- Built Flask-based RESTful microservices integrated with React dashboards to deliver real-time sensor insights and ML model outputs to end users; containerized with Docker for reproducible deployment.

INDUSTRY EXPERIENCE

AI/ML Engineer Intern | PNC Financial Services, USA

Jan 2024 – Jul 2024 • Remote, USA

- Built end-to-end ML pipelines using AWS SageMaker and Airflow for financial risk systems, improving processing efficiency by 30%.
- Developed an ensemble fraud scoring system combining XGBoost on structured transaction data with a fine-tuned LLM on unstructured data, reducing detection latency by 15%.
- Designed and deployed scalable ML services via Java Spring Boot microservices, Docker, and Kubernetes across cloud environments, ensuring high availability and seamless enterprise integration.

Senior Software Developer | ML & Backend | Tata Consultancy Services

Jul 2021 – Apr 2023 • India

- Developed and deployed ML models for predictive analytics on the Mars Petcare platform, improving demand forecasting accuracy by 25% using gradient boosting and time-series modeling.
- Implemented OWASP-compliant secure backend systems for enterprise applications, improving performance by 20% and reducing security vulnerabilities by 35%.
- Deployed and maintained scalable cloud infrastructure on AWS (EC2, S3, Lambda) supporting high availability for production ML services.

Senior Software Engineer | Foray Software Private Limited

Jul 2019 – May 2021 • India

- Enhanced front-end inventory search functionality for Mars employees, delivering 30% faster search experience and 15% increase in task efficiency.
- Partnered with cross-functional teams to integrate features and ensure seamless system functionality, improving overall user satisfaction scores by 20%

Technologies: *Drupal, JavaScript, HTML/CSS, PHP, MySQL*

Associate Analyst | PurpleTalk

Feb 2017 – Jun 2019 • India

- Built peer-to-peer payment features for a consumer Android app (Canya) and the payment and invoice module for an enterprise attendance management platform (SILA), handling transaction flows and billing logic end to end.
- Developed and unit-tested modules across the Upshot user engagement platform and a social networking alarm app (Cuckuu); received Best Employee and Best Performance awards for delivery across concurrent projects.

Technologies: *Yii Framework, MongoDB, JavaScript, jQuery*

Research Associate | Excelra

Jun 2016 – Dec 2016 • India

- Built an automated visualization tool mapping disease associations and biological pathways for a target gene ID, improving bioinformatics workflow efficiency by 25%.

KEY PROJECTS

AI Companion for Seniors | [Ongoing] | Anthropic Claude API, AWS Bedrock, Agentic RAG, MongoDB, FastAPI, React, Python

- Voice-enabled multimodal health companion built on the Anthropic Claude API and AWS Bedrock, using the same service-oriented, component-swappable RAG design as my published research.
- Extracts structured medical data from conversation and documents to drive medication reminders and anomaly detection on vitals, surfaced through a real-time care dashboard; extending clinician access with a care team.

Weather Impact on Music Trends | Python, PostgreSQL, Selenium, Essentia, Scikit-learn, Plotly, Docker

- Built an ETL pipeline across 103 cities joining Selenium-scraped Apple Music charts with OpenWeather data in a PostgreSQL schema; extracted BPM, energy, and valence from audio previews using Essentia in Docker.
- Applied PCA and KMeans clustering to group cities by listening profile, surfacing genre and tempo patterns tied to weather conditions.

Strawberry Fruit Ripeness Detection | Python, Scikit-learn, Pandas, Vercel

- KNN + SVM ensemble classifying strawberry ripeness from images at 92% accuracy, deployed as a Vercel web app returning predictions from uploaded photos.